**20 SQL routines used on the Sick database**

1) Display the count of the number of rows in the Patient table.

```
--This command counts the number of records in the Patient table and lists them as PatientCount
Select Count(*) as PatientCount from Patient
```

98 % 1 0

Results | Messages

| | PatientCount |
|---|---|
| 1 | 27 |

2) Delete a specified record in the ProviderType table and prove that it is deleted. (before and after)

```sql
--Deletes the provider
Select * from ProviderType
DELETE from ProviderType WHERE ProviderTypeID = 5
Select * from ProviderType
```

98 % No issues found

Results Messages

| | ProviderTypeID | ProviderType | ProTypeDesc |
|---|---|---|---|
| 1 | 1 | Physician | MD/DO |
| 2 | 2 | Registered Nurse | RN |
| 3 | 3 | Physical Therapist | NULL |
| 4 | 4 | Phlebotomist | Blood Draw |
| 5 | 5 | Dermotologist | DERM |

| | ProviderTypeID | ProviderType | ProTypeDesc |
|---|---|---|---|
| 1 | 1 | Physician | MD/DO |
| 2 | 2 | Registered Nurse | RN |
| 3 | 3 | Physical Therapist | NULL |
| 4 | 4 | Phlebotomist | Blood Draw |

3) Two table implied join qualified with a key value.

```sql
-- This code uses the Provider and ProviderType table to compile Provider names and their profession
Select FName, LName, ProviderType
From Provider, ProviderType
Where Provider.ProviderTypeID = ProviderType.ProviderTypeID
```

98 % No issues found

Results Messages

| | FName | LName | ProviderType |
|---|---|---|---|
| 1 | James | Wilson | Physician |
| 2 | Emily | Carter | Registered Nurse |
| 3 | Michael | Reed | Physician |
| 4 | Sarah | Nguyen | Physical Therapist |
| 5 | David | Patel | Physician |
| 6 | Laura | Hernandez | Registered Nurse |
| 7 | Robert | Kim | Phlebotomist |
| 8 | Jessica | Morris | Physical Therapist |
| 9 | Daniel | Lopez | Physician |
| 10 | Amanda | Brooks | Registered Nurse |
| 11 | Matthew | Sanders | Physician |
| 12 | Olivia | Price | Physical Therapist |
| 13 | Andrew | Bell | Phlebotomist |
| 14 | Stephanie | Rivera | Registered Nurse |
| 15 | Christopher | Howard | Physician |
| 16 | Nicole | Ward | Physical Therapist |
| 17 | Brian | Cox | Physician |
| 18 | Rachel | Peterson | Registered Nurse |
| 19 | Kevin | Flores | Phlebotomist |
| 20 | Megan | Russell | Physical Therapist |
| 21 | Jonathan | Long | Physician |
| 22 | Hannah | Ramirez | Registered Nurse |
| 23 | Eric | Powell | Phlebotomist |
| 24 | Victoria | Bennett | Physical Therapist |
| 25 | Steven | Coleman | Physician |

4) Three table implied join qualified with a key value.

```sql
-- Uses the Patient, Ailment, and PatientAilment table to gather patient names, ailments, and their start and end date.
Select P.FName, P.LName, Ailment, StartDate, EndDate
From Patient P, Ailment A, PatientAilment PA
Where P.PatientID = PA.PatientID
and A.AilmentID = PA.AilmentID
```

98 % No issues found

Results Messages

| | FName | LName | Ailment | StartDate | EndDate |
|---|---|---|---|---|---|
| 1 | John | Smith | Hypertension | 2018-01-10 | NULL |
| 2 | John | Smith | Migraine | 2020-06-15 | 2020-09-01 |
| 3 | Mary | Johnson | Diabetes Mellitus | 2019-03-22 | NULL |
| 4 | Mary | Johnson | Seasonal Allergies | 2021-04-01 | 2021-05-15 |
| 5 | Robert | Brown | Asthma | 2017-08-12 | NULL |
| 6 | Robert | Brown | Chronic Back Pain | 2022-02-10 | NULL |
| 7 | Patricia | Taylor | Depression | 2020-11-05 | NULL |
| 8 | Patricia | Taylor | Eczema | 2019-01-14 | 2019-06-30 |
| 9 | Michael | Anderson | Arthritis | 2016-05-19 | NULL |
| 10 | Michael | Anderson | Irritable Bowel Syndrome | 2021-07-07 | NULL |
| 11 | Linda | Thomas | Anxiety Disorder | 2018-09-03 | NULL |
| 12 | Linda | Thomas | Vertigo | 2020-10-10 | 2020-12-01 |
| 13 | William | Jackson | Coronary Artery Disease | 2015-02-17 | NULL |
| 14 | William | Jackson | Chronic Obstructive Pulmonary Disease | 2022-01-01 | NULL |
| 15 | Barbara | White | Seasonal Allergies | 2019-04-20 | 2019-06-01 |
| 16 | Barbara | White | Influenza | 2023-01-05 | 2023-01-20 |
| 17 | James | Harris | Migraine | 2018-07-11 | NULL |
| 18 | James | Harris | Sleep Apnea | 2021-09-09 | NULL |
| 19 | Susan | Martin | Hypertension | 2016-03-30 | NULL |
| 20 | Susan | Martin | Anemia | 2022-05-14 | NULL |
| 21 | David | Thomp... | Pneumonia | 2020-12-01 | 2020-12-20 |
| 22 | David | Thomp... | Urinary Tract Infection | 2023-02-02 | 2023-02-10 |
| 23 | Karen | Garcia | Gastroesophageal Reflux Disease | 2019-06-18 | NULL |
| 24 | Karen | Garcia | Osteoporosis | 2021-08-25 | NULL |
| 25 | Richard | Martinez | Chronic Kidney Disease | 2017-01-09 | NULL |
| 26 | Richard | Martinez | Hypothyroidism | 2022-04-04 | NULL |

5) Use twelve SQL functions (such as SUM).

```sql
--1. Counts the number of Patient records as PatientCount
Select Count(*) as PatientCount from Patient

--2. Averages out the PatientID as AVGPID
Select AVG(PatientID) as AVGPID from Patient

--3. Calculates the maximum PatientID as MAXPID
Select MAX(PatientID) as MAXPID from Patient

--4. Calculates the minimum PatientID as MINPID
Select MIN(PatientID) as MINPID from Patient

--5. Creates a table showing the start and end date of all patient's ailments and calculates the length of sickness
Select StartDate, EndDate, DATEDIFF(day, StartDate, EndDate) as Length
From PatientAilment

--6. Adds the PatientID as SUMPID
Select SUM(PatientID) as SUMPID from Patient

--7. Calculates the length of each patients last name
Select LEN(LName) as LengthLastName from Patient

--8. Changes all letters in a patients last name to uppercase
Select upper(LName) as UpperLastName from Patient

--9. Changes all letters in a patients last name to lowercase
Select lower(LName) as LowerLastName from Patient

--10. Reverses a patients first name
Select Reverse(FName) as LowerLastName from Patient

--11. Calculates the Logarithm of each PatientID
Select LOG(PatientID) as Logarithn from Patient

--12. Returns each PatientID squared
Select Square(PatientID) as PIDSquared from Patient
```

| | PatientCount |
|---|---|
| 1 | 27 |

| | AVGPID |
|---|---|
| 1 | 14 |

| | MAXPID |
|---|---|
| 1 | 27 |

| | MINPID |
|---|---|
| 1 | 1 |

| | StartDate | EndDate | Length |
|---|---|---|---|
| 1 | 2018-01-10 | NULL | NULL |
| 2 | 2020-06-15 | 2020-09-01 | 78 |
| 3 | 2019-03-22 | NULL | NULL |
| 4 | 2021-04-01 | 2021-05-15 | 44 |
| 5 | 2017-08-12 | NULL | NULL |
| 6 | 2022-02-10 | NULL | NULL |
| 7 | 2020-11-05 | NULL | NULL |
| 8 | 2019-01-14 | 2019-06-30 | 167 |

| | SUMPID |
|---|---|
| 1 | 378 |

| | LengthLastName |
|---|---|
| 1 | 5 |
| 2 | 7 |
| 3 | 5 |
| 4 | 6 |
| 5 | 8 |
| 6 | 6 |
| 7 | 7 |
| 8 | 5 |

| | UpperLastName |
|---|---|
| 1 | SMITH |
| 2 | JOHNSON |
| 3 | BROWN |
| 4 | TAYLOR |
| 5 | ANDERSON |
| 6 | THOMAS |
| 7 | JACKSON |
| 8 | WHITE |

| | LowerLastName |
|---|---|
| 1 | smith |
| 2 | johnson |
| 3 | brown |
| 4 | taylor |
| 5 | anderson |
| 6 | thomas |
| 7 | jackson |
| 8 | white |

| | LowerLastName |
|---|---|
| 1 | nhoJ |
| 2 | yraM |
| 3 | treboR |
| 4 | aicirtaP |
| 5 | leahciM |
| 6 | adniL |
| 7 | mailliW |
| 8 | arabraB |

| | Logarithn |
|---|---|
| 1 | 0 |
| 2 | 0.693147180559945 |
| 3 | 1.09861228866811 |
| 4 | 1.38629436111989 |
| 5 | 1.6094379124341 |
| 6 | 1.79175946922805 |
| 7 | 1.94591014905531 |
| 8 | 2.07944154167984 |

| | PIDSquared |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |
| 4 | 16 |
| 5 | 25 |
| 6 | 36 |
| 7 | 49 |
| 8 | 64 |

6) Write an SQL statement that utilizes 'like' in the where clause.

```sql
--This statement uses like to find all patients that have their ailment start with 'Hypo'
Select P.FName, P.LName, Ailment, StartDate, EndDate
From Patient P, Ailment A, PatientAilment PA
Where P.PatientID = PA.PatientID
and A.Ailment like 'Hypo%'
```

98 % No issues found Ln: 1, Ch: 3 TABS

Results Messages

| | FName | LName | Ailment | StartDate | EndDate |
|---|---|---|---|---|---|
| 1 | John | Smith | Hypothyroidism | 2018-01-10 | NULL |
| 2 | John | Smith | Hypothyroidism | 2020-06-15 | 2020-09-01 |
| 3 | Mary | Johnson | Hypothyroidism | 2019-03-22 | NULL |
| 4 | Mary | Johnson | Hypothyroidism | 2021-04-01 | 2021-05-15 |
| 5 | Robert | Brown | Hypothyroidism | 2017-08-12 | NULL |
| 6 | Robert | Brown | Hypothyroidism | 2022-02-10 | NULL |
| 7 | Patricia | Taylor | Hypothyroidism | 2020-11-05 | NULL |
| 8 | Patricia | Taylor | Hypothyroidism | 2019-01-14 | 2019-06-30 |
| 9 | Michael | Anderson | Hypothyroidism | 2016-05-19 | NULL |
| 10 | Michael | Anderson | Hypothyroidism | 2021-07-07 | NULL |
| 11 | Linda | Thomas | Hypothyroidism | 2018-09-03 | NULL |
| 12 | Linda | Thomas | Hypothyroidism | 2020-10-10 | 2020-12-01 |

7) Demonstrate the use of the rowcount feature to return 4 rows from a table that has more than 4 rows. Then demonstrate the use of the rowcount feature to reset the number of rows returned to all of the rows in a table. Demonstrate an alternative to rowcount that has similar functionality.

```sql
--Only shows the top 4 rows from the Ailment table
set rowcount 4
select * from Ailment

--in order to show all of the rows after using rowcount you have to do 'set rowcount 0'
```

98 % No issues found

Results Messages

| | AilmentID | Ailment | AilDesc |
|---|---|---|---|
| 1 | 1 | Hypertension | Persistently elevated blood pressure requiring monitori... |
| 2 | 2 | Diabetes Mellitus | Metabolic disorder characterized by high blood glucos... |
| 3 | 3 | Asthma | Chronic respiratory condition causing airway inflammati... |
| 4 | 4 | Migraine | Recurring headaches often accompanied by nausea ... |

8) Create a SQL routine to insert values into the Ailment table.

```sql
Select * from Ailment
--This command adds the broken leg ailment to this database and gives it the AilmentID of 26
Insert into Ailment --insert statement
    (AilmentID, Ailment, AilDesc)
Values
    (26, 'Broken Leg', 'Broken bones')

Select * from Ailment
```

98 % No issues found

Results Messages

| | AilmentID | Ailment | AilDesc |
|---|---|---|---|
| 18 | 18 | Sleep Apnea | Sleep disorder characterized by repeated pauses in br... |
| 19 | 19 | Anemia | Condition where lack of healthy red blood cells reduce... |
| 20 | 20 | Hypothyroidism | Underactive thyroid gland leading to fatigue and weig... |
| 21 | 21 | Hyperthyroidism | Overactive thyroid gland causing weight loss and rapi... |
| 22 | 22 | Urinary Tract Infection | Infection affecting any part of the urinary system. |
| 23 | 23 | Chronic Obstructive Pulmonary ... | Progressive lung disease causing breathing difficulty. |
| 24 | 24 | Vertigo | Sensation of spinning or dizziness often related to inne... |
| 25 | 25 | Peptic Ulcer Disease | Open sores developing on the inner lining of the stom... |

| | AilmentID | Ailment | AilDesc |
|---|---|---|---|
| 17 | 17 | Osteoporosis | Condition marked by weakened bones and incre... |
| 18 | 18 | Sleep Apnea | Sleep disorder characterized by repeated pause... |
| 19 | 19 | Anemia | Condition where lack of healthy red blood cells r... |
| 20 | 20 | Hypothyroidism | Underactive thyroid gland leading to fatigue and... |
| 21 | 21 | Hyperthyroidism | Overactive thyroid gland causing weight loss an... |
| 22 | 22 | Urinary Tract I... | Infection affecting any part of the urinary system. |
| 23 | 23 | Chronic Obstru... | Progressive lung disease causing breathing diffi... |
| 24 | 24 | Vertigo | Sensation of spinning or dizziness often related t... |
| 25 | 25 | Peptic Ulcer Di... | Open sores developing on the inner lining of the ... |
| 26 | 26 | Broken Leg | Broken bones |

9) Create a SQL routine to update one middle name in the Patient table.

```sql
Select * from Patient Where PatientID = 5
go
--This command updates the patient with the PatientID 5 and gives him a middle name
Update Patient --table
    Set MName = 'T'
Where PatientID = 5
go

Select * from Patient Where PatientID = 5
```

98 % 1 0

Results | Messages

| | PatientID | FName | LName | MName | DOB | Gender | Height | Weight |
|---|---|---|---|---|---|---|---|---|
| 1 | 5 | Michael | Anderson | NULL | 1995-05-09 | M | 69.0 | 175.0 |

| | PatientID | FName | LName | MName | DOB | Gender | Height | Weight |
|---|---|---|---|---|---|---|---|---|
| 1 | 5 | Michael | Anderson | T | 1995-05-09 | M | 69.0 | 175.0 |

10) Create SQL to truncate and populate a new table with at least 10 rows using data you supply in the insert statement. Table must have at least 3 columns and is not a table in the Sick database.

```sql
Truncate table Person --Deletes records in table
go

--Adds records into the Person table
Insert into Person
    (FName, LName)
Values
    ('Jim', 'Johnson'),
    ('Steve', 'Smith'),
    ('Sara', 'White')
go
Select * from Person
```

98 % 1 0

Results | Messages

| | PersonID | FName | LName |
|---|---|---|---|
| 1 | 1 | Jim | Johnson |
| 2 | 2 | Steve | Smith |
| 3 | 3 | Sara | White |

11) Create a SQL statement that inserts 2 Ailments for a Patient of your choice and 3 Ailments for another Patient of your choice.

```sql
Select * from PatientAilment
--GIves two different Patients different Ailments
Insert into PatientAilment
    (PatientID, AilmentID)
Values
    (25,3),
    (25,4),
    (26,3),
    (26,4),
    (26,6)
Select * from PatientAilment
```

98 % No issues found

Results | Messages

| | | | | | |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 2018-01-10 | NULL | 1 |
| 2 | 1 | 4 | 2020-06-15 | 2020-09-01 | 3 |
| 3 | 2 | 2 | 2019-03-22 | NULL | 2 |
| 4 | 2 | 10 | 2021-04-01 | 2021-05-15 | 5 |
| 5 | 3 | 3 | 2017-08-12 | NULL | 4 |
| 6 | 3 | 9 | 2022-02-10 | NULL | 6 |
| 7 | 4 | 6 | 2020-11-05 | NULL | 7 |
| 8 | 4 | 15 | 2019-01-14 | 2019-06-30 | 8 |

| | PatientID | AilmentID | StartDate | EndDate | ProviderID |
|---|---|---|---|---|---|
| 50 | 25 | 4 | NULL | NULL | NULL |
| 51 | 25 | 19 | 2023-03-03 | NULL | 22 |
| 52 | 26 | 1 | NULL | NULL | NULL |
| 53 | 26 | 2 | NULL | NULL | NULL |
| 54 | 26 | 3 | NULL | NULL | NULL |
| 55 | 26 | 4 | NULL | NULL | NULL |
| 56 | 26 | 5 | NULL | NULL | NULL |
| 57 | 26 | 6 | NULL | NULL | NULL |

12) Create a query that uses Group By with the  Patient and Ailment table.

```sql
--Orders the patients by the number of ailments they have
Select FName, LName, Count(AilmentID) as AilmentCount
From Patient P, PatientAilment PA
Where P.PatientID = PA.PatientID
Group By FName, LName
Order By AilmentCount Desc
```

98 %  No issues found

Results  Messages

| | FName | LName | AilmentCount |
|---|---|---|---|
| 1 | William | Williams | 6 |
| 2 | Mark | Lopez | 5 |
| 3 | Susan | Martin | 2 |
| 4 | Richard | Martinez | 2 |
| 5 | Nancy | Robinson | 2 |
| 6 | Lisa | Rodriguez | 2 |
| 7 | John | Smith | 2 |
| 8 | Patricia | Taylor | 2 |
| 9 | Linda | Thomas | 2 |
| 10 | David | Thompson | 2 |
| 11 | Charles | Walker | 2 |
| 12 | Barbara | White | 2 |
| 13 | Karen | Garcia | 2 |
| 14 | Sandra | Hall | 2 |
| 15 | James | Harris | 2 |
| 16 | William | Jackson | 2 |

13) Create a SQL statement that utilizes the substring command.

```sql
--Shows part of every patients last name only using the 1st through the third character
Select Substring(LName, 1, 3) as 'Partial Last Name'  --Field Name, Start Char, End Char
From Patient
```

98 % No issues found

Results Messages

| | Partial Last Name |
|---|---|
| 1 | Smi |
| 2 | Joh |
| 3 | Bro |
| 4 | Tay |
| 5 | And |
| 6 | Tho |
| 7 | Jac |
| 8 | Whi |
| 9 | Har |
| 10 | Mar |
| 11 | Tho |
| 12 | Gar |
| 13 | Mar |
| 14 | Rob |
| 15 | Cla |
| 16 | Rod |

14) Demonstrate the use of distinct, ascending, descending in a query.

```sql
--Orders last name in ascending order
Select Distinct LName from Patient
Order by LName Asc
Go
--Orders last name in descending order
Select Distinct LName from Patient
Order by LName Desc
```

98 % 1 0

Results | Messages

| | LName |
|---|---|
| 1 | Allen |
| 2 | Anderson |
| 3 | Brown |
| 4 | Clark |
| 5 | Garcia |
| 6 | Hall |
| 7 | Harris |
| 8 | Jackson |

| | LName |
|---|---|
| 1 | Young |
| 2 | Wright |
| 3 | Williams |
| 4 | White |
| 5 | Walker |
| 6 | Thom... |
| 7 | Thomas |

15) Demonstrate @@version

```
--Displays the version of SQL Server
Select @@Version
```

98 %  1  0

Results | Messages

| | (No column name) |
|---|---|
| 1 | Microsoft SQL Server 2025 (RTM-GDR) (KB5073177... |

16) How many Ailments are in Ailment table (number, not the list of records).

```
--Shows the total number of ailments in the database
Select Count(*) as AilmentCount from Ailment
```

98 %

No issues found

Results | Messages

| | AilmentCount |
|---|---|
| 1 | 26 |

18) Which Patient had the most Ailments.

```sql
--Orders the Patients by their number of ailments
Select FName, LName, Count(AilmentID) as AilmentCount
From Patient P, PatientAilment PA
Where P.PatientID = PA.PatientID
Group By FName, LName
Order By AilmentCount Desc
```

98 % No issues found

Results Messages

| | FName | LName | AilmentCount |
|---|---|---|---|
| 1 | William | Williams | 6 |
| 2 | Mark | Lopez | 5 |
| 3 | Susan | Martin | 2 |
| 4 | Richard | Martinez | 2 |
| 5 | Nancy | Robinson | 2 |
| 6 | Lisa | Rodriguez | 2 |
| 7 | John | Smith | 2 |
| 8 | Patricia | Taylor | 2 |
| 9 | Linda | Thomas | 2 |
| 10 | David | Thompson | 2 |
| 11 | Charles | Walker | 2 |
| 12 | Barbara | White | 2 |

19) Which Provider member had the longest Last Name.

```sql
--Orders the providers by the length of their last name
Select LName, Len(LName) as LNameLength
From Provider
Order By LNameLength Desc
```

98 % No issues found

Results Messages

| | LName | LNameLength |
|---|---|---|
| 1 | Hernandez | 9 |
| 2 | Peterson | 8 |
| 3 | Sanders | 7 |
| 4 | Russell | 7 |
| 5 | Ramirez | 7 |
| 6 | Bennett | 7 |
| 7 | Coleman | 7 |
| 8 | Brooks | 6 |
| 9 | Powell | 6 |
| 10 | Flores | 6 |
| 11 | Rivera | 6 |
| 12 | Howard | 6 |
| 13 | Morris | 6 |
| 14 | Wilson | 6 |
| 15 | Carter | 6 |
| 16 | Nguyen | 6 |

20) Write a script that creates the data to have a Patient with more than one Ailment, as well as a script to display that Patient and their Ailments.

```sql
--Creating the Patient
Insert into Patient
    (PatientID, FName, LName, DOB)
Values
    (28,'Wyatt', 'Karnes', '2006-12-20')
go

--Giving the patient ailments
--Wyatt Karnes PatientID is 28
-- 3-Asthma, 19-Anemia
Insert into PatientAilment
    (PatientID, AilmentID)
Values
    (28,3),
    (28,19)

Select P.FName, P.LName, Ailment
From Patient P, Ailment A, PatientAilment PA
Where P.PatientID = PA.PatientID
and A.AilmentID = PA.AilmentID
and P.PatientID = 28
```

98 % No issues found

Results Messages

| | FName | LName | Ailment |
|---|---|---|---|
| 1 | Wyatt | Karnes | Asthma |
| 2 | Wyatt | Karnes | Anemia |